
Shunting Inhibition and Dendritic Branching Shape Local Credit Assignment

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Abstract

1 Biological neurons assign credit through structures that standard neural networks
2 usually ignore: synapses are distributed across branching dendrites, excitatory
3 and inhibitory inputs evoke synaptic conductances, and inhibition can shunt local
4 voltage by changing total input resistance. We study conductance-based dendritic
5 networks with E/I synapse banks, shunting inhibition, and tree-structured branch-
6 to-soma coupling, and ask when a low-bandwidth somatic broadcast can replace
7 compartment-specific backpropagated errors. Exact gradients for these dendritic
8 trees show that each synaptic update separates into a local eligibility term built
9 from presynaptic activity, driving force, and input resistance, and one non-local
10 compartment error. Under identity upward transfer, that error is the somatic error
11 transported through conductance-stage path gains. This factorization turns local
12 learning into a credit-signal compression problem: broadcast is useful only when it
13 approximates the path-specific compartment-error field. We test this mechanism
14 across gradient, oracle-transport, inhibition-intervention, competence, morphology,
15 and harder-data diagnostics. Shunting improves LocalCA to backprop alignment,
16 reduces gradient-scale mismatch, and makes exact compartment errors more rank-1
17 compressible than matched additive controls in the regimes studied here. Inhibitory-
18 conductance interventions show that learned inhibition carries sample-specific
19 pathway structure, while transported-error oracle experiments show that restoring
20 the exact effective path-gain error largely closes the local-learning gap. Under
21 nonnegative conductances and rank-1 broadcast, 5F shunting LocalCA remains
22 about 5 percentage points below matched backpropagation on MNIST, Fashion-
23 MNIST, and context gating. These results suggest that E/I conductance, shunting
24 inhibition, and dendritic branching can reshape credit-signal geometry and make
25 low-bandwidth local learning more faithful.

1 Introduction

27 Unlike point units in standard neural networks, biological neurons assign credit across branching
28 dendritic trees. A synapse does not only see presynaptic activity and a scalar postsynaptic activation;
29 it also has access to local membrane voltage, synaptic reversal potential, total conductance, and input
30 resistance. Inhibitory synapses can therefore do more than subtract activity: by increasing total local
31 conductance, inhibition can shunt voltage and change the gain of every dendritic path passing through
32 that compartment.

33 We examine whether these biophysical ingredients can facilitate local credit assignment (LocalCA).
34 Backpropagation [15] would assign a distinct error to every dendritic compartment, whereas a
35 biologically plausible supervisory signal is more likely to arrive as a low-bandwidth somatic or
36 modulatory broadcast [20]. The central question is therefore not whether such a broadcast can
37 reproduce unconstrained backpropagation in every setting, but when the exact dendritic error field
38 is simple enough for it to approximate. Our contribution is not simply to add dendritic structure to

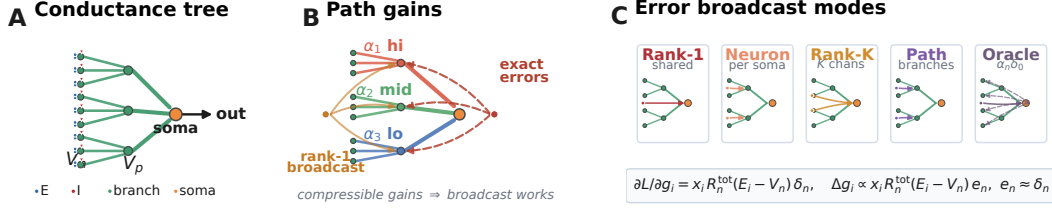


Figure 1: **Model and credit assignment.** (A) Conductance tree with nonnegative excitatory drive and inhibitory shunting load. (B) Exact credit varies by dendritic path, whereas rank-1 LocalCA broadcasts one shared error field. (C) LocalCA preserves the exact synapse-local eligibility and substitutes $e_n \approx \delta_n$; rank-1, neuron-wise, rank- K , and path-structured broadcasts trade biological bandwidth for fidelity, while transported error is an oracle.

a neural network, but to show how conductance-based synapses, shunting inhibition, and dendritic topology change the geometry of the credit signal itself.

Starting from conductance-based dendritic voltage equations [1], we derive exact gradients for dendritic trees (Theorem 1). Each synaptic gradient factorizes into a synapse-local eligibility term and a single non-local compartment error. The eligibility term depends on presynaptic activity, driving force, and input resistance. The compartment error is path-specific: under identity upward transfer it is the somatic error multiplied by a conductance-stage path gain through the dendritic tree. LocalCA preserves the exact local eligibility and replaces this path-specific error with a broadcast field.

This factorization turns local credit assignment into a compressibility problem. By compressibility, we mean that the matrix of exact compartment errors across examples and compartments is well approximated by a low-rank signal, especially a rank-1 field shared across compartments. Dendritic branching creates the paths; E/I synapse banks set the local conductance state; and shunting inhibition changes the input resistance along each path. Shunting is not assumed to help monotonically. It helps rank-1 broadcast when inhibitory conductance attenuates high-gain paths or otherwise makes the exact error field more compressible.

Related work. This work builds on models of dendritic signal integration, branch-level nonlinear computation, divisive normalization, and biologically plausible teaching signals [1, 3, 4, 8, 9, 17–19, 22, 26–28, 34, 35, 45]. It also connects to machine-learning approaches that replace exact backpropagation with local, random, predictive, equilibrium, perturbation, or broadcast feedback signals [13–16, 21, 29–33, 36–40]. Our contribution is complementary: we derive the exact conductance-tree credit object and test when shunting inhibition and dendritic morphology make that object broadcast-compatible.

We test this chain directly. Exact-gradient reconstruction verifies the factorization numerically. Gradient-fidelity diagnostics show that shunting produces local updates that are better aligned and better scaled relative to backpropagation than matched additive controls. Exact-error compressibility and inhibition-intervention diagnostics show that learned inhibitory conductance carries sample-specific pathway structure. Transported-error oracle experiments show that supplying the exact effective path-gain-transported compartment error largely closes the local-learning gap, identifying broadcast fidelity as an important bottleneck. In an MNIST gradient-dynamics diagnostic, shunting improves final LocalCA to backprop alignment (cosine 0.100 ± 0.033 vs. 0.005 ± 0.026) and reduces scale mismatch (0.26 ± 0.05 vs. 0.50 ± 0.05). On noise resilience under matched inhibition (at least five inhibitory synapses per branch), transported error raises additive local learning from 81–84% under rank-1 broadcast to 92–94%, nearly matching shunting at 95–95.2%.

Contributions. This paper makes three contributions: deriving the conductance-stage path gain, its local-eligibility/path-error factorization, and a closed-form inhibitory path-gain ratio; identifying path-gain compressibility as the quantity mediating whether low-rank somatic broadcast can approximate exact compartment errors; and showing empirically that shunting inhibition can improve this approximation while also identifying regimes where rank-1 feedback fails.

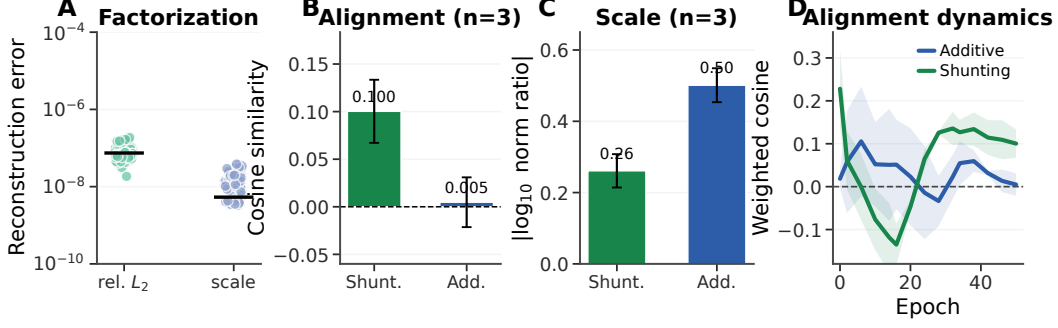


Figure 2: **Gradient-fidelity diagnostic.** (A) Exact-factorization sanity check: reconstructed gradients match autograd to numerical precision. (B) Final-state weighted cosine similarity between local and backprop gradients. (C) Final-state scale mismatch, $|\log_{10}(\|g^{\text{local}}\|/\|g^{\text{bp}}\|)|$. Error bars are ± 1 s.d. across 3 gradient-dynamics seeds; main learning and oracle sweeps use 5 seeds unless noted. (D) Alignment over learning; Appendix Fig. S3 shows finite gradient norms.

2 Compartmental Voltage Model and Gradient Derivation

For each dendritic compartment, we use a steady-state conductance model derived from discretized passive cable dynamics [1, 2]. In normalized units, leak and inhibitory reversal potentials are set to 0, the excitatory reversal potential is set to 1, and leak conductance is fixed at 1 (i.e., all conductances are expressed relative to leak). Each compartment computes a conductance-weighted voltage: excitatory conductance pulls the voltage toward the excitatory reversal potential, leak and shunting inhibition pull it toward zero, and dendritic coupling transfers child voltages upward toward the soma. This form makes local sensitivity depend on both driving force ($E - V$) and total input resistance R^{tot} . Throughout the paper we write dendritic morphologies as rooted trees $[b_1, b_2, \dots, b_D]$, where each factor gives the fan-in from one dendritic stage to the next on the path to the soma. Thus $[3, 3]$ means that each soma aggregates 3 proximal branches and each proximal branch aggregates 3 distal branches, for 9 distal leaves per soma. In the main feedforward experiments, synaptic inputs terminate on dendritic branches rather than directly on the soma. Each branch layer carries explicit excitatory conductance-based synapse banks and, when enabled, learned inhibitory branch-level conductance-based synapse banks driven by nonnegative inputs. This is input-driven shunting inhibition, not a recurrent lateral inhibitory circuit; explicit inhibitory-cell controls are reported in the appendix. The additive and shunting cores are architecture-matched at the tree level and differ only in the branch-level synaptic integration rule. These ingredients play distinct roles in the credit-assignment argument. Branching creates the path structure. E/I synapse banks determine the local conductance state. Shunting turns inhibitory conductance into a multiplicative change in path gain through input resistance. The local learning rule then asks whether a low-bandwidth teaching signal can stand in for the exact path-specific error induced by that tree. **Notation.** We use α_n^{cond} for the conductance-stage path gain from compartment n to the soma and $\tilde{\alpha}_n$ for the effective path gain after including any post-voltage activation derivatives; when upward transfer is identity, $\tilde{\alpha}_n = \alpha_n^{\text{cond}}$. We use N_E and N_I for the number of excitatory and inhibitory synapses per branch, r_n^{4F} for the empirical 4F branch-soma covariance proxy, ϕ_n for the 5F confidence factor, λ_{mix} for the pathway-broadcast mixing weight, and κ_j for depth-dependent modulation in morphology extensions. All conductances and presynaptic activities in the main experimental setting are nonnegative; additive voltages may be signed comparators, but the conductance variables that define the shunting model are nonnegative.

Voltage equation and local sensitivities. Consider compartment n with synaptic inputs j (activity x_j , reversal E_j , conductance $g_j^{\text{syn}} \geq 0$) and dendritic inputs from children (voltage V_j , conductance $g_j^{\text{den}} \geq 0$). The steady-state voltage is:

$$V_n = \frac{\sum_j E_j x_j g_j^{\text{syn}} + \sum_j V_j g_j^{\text{den}}}{\underbrace{\sum_j x_j g_j^{\text{syn}} + \sum_j g_j^{\text{den}} + 1}_{g_n^{\text{tot}}}}, \quad R_n^{\text{tot}} = 1/g_n^{\text{tot}}. \quad (1)$$

109 V_n is a convex combination of reversal potentials, child voltages, and leak. Writing $\mathcal{S}_n = \{0\} \cup$
 110 $\{E_j\}_j \cup \{V_j\}_j$, we have $\min \mathcal{S}_n \leq V_n \leq \max \mathcal{S}_n$ and $0 < R_n^{\text{tot}} \leq 1$. The local sensitivities follow
 111 directly:

Proposition 1 (Local Sensitivities).

$$\frac{\partial V_n}{\partial g_i^{\text{syn}}} = x_i R_n^{\text{tot}} (E_i - V_n), \quad \frac{\partial V_n}{\partial V_i} = g_i^{\text{den}} R_n^{\text{tot}}, \quad \frac{\partial V_n}{\partial g_i^{\text{den}}} = R_n^{\text{tot}} (V_i - V_n). \quad (2)$$

112 Eq. (2) gives the eligibility factors used by the local rules below: presynaptic activity or voltage
 113 difference, driving force, and input resistance.

114 **Shunting inhibition as divisive gain control.** An inhibitory synapse with $E_{\text{inh}} \approx 0$ contributes
 115 current $(0 - V_n)x_j g_j^{\text{syn}}$ and increases g_n^{tot} . Its sensitivity is $\partial V_n / \partial g_j^{\text{syn}} = -x_j R_n^{\text{tot}} V_n$, which corre-
 116 sponds to multiplicative attenuation (divisive normalization). Shunting is divisive at the voltage level,
 117 but its effect on firing rates can be subtractive in some regimes [10]. Inhibitory plasticity can balance
 118 excitation dynamically [11]; our learned inhibitory conductances serve an analogous role. The same
 119 denominator also gives the direct connection to path gain. Writing $G_n^E(x)$ and $G_n^I(x)$ for the total
 120 excitatory and inhibitory synaptic conductance impinging on compartment n ,

$$R_n^{\text{tot}}(x) = (1 + G_n^E(x) + G_n^I(x) + G_n^{\text{den}})^{-1}.$$

121 Increasing $G_n^I(x)$ lowers $R_n^{\text{tot}}(x)$ and therefore multiplicatively suppresses every upstream path
 122 whose error must pass through compartment n . In this sense inhibition can gate credit flow by
 123 changing the path gain, even when the inhibitory drive is feedforward rather than lateral. Prop. 2
 124 formalizes this statement after the tree path gain is defined.

125 **Exact gradients for dendritic trees.** Let V_0 be the somatic output and define the soma/core error
 126 as $\delta_0 := \partial L / \partial V_0$. For a linear decoder $\hat{y} = W_{\text{dec}} V_0$, this reduces to $\delta_0 = W_{\text{dec}}^T (\partial L / \partial \hat{y})$; for a
 127 nonlinear decoder, δ_0 is the corresponding decoder-input Jacobian product. The theoretical derivation
 128 is stated for the conductance-stage voltage before any post-voltage reactivation. When a branch
 129 applies a monotone reactivation before transmitting activity upward, the same recursion holds with
 130 additional local derivative factors along the path. All exact-error diagnostics below are computed as
 131 pre-activation voltage errors, including activation-derivative factors when reactivation is enabled.

132 **Theorem 1** (Conductance-Stage Backpropagation on a Dendritic Tree). *For a rooted dendritic tree*
 133 *with soma at node 0 and unique parent $p(n)$ for each non-somatic compartment, suppose the parent*
 134 *receives either the pre-activation voltage V_n or a local post-voltage activity $a_n = f_n(V_n)$. For*
 135 *identity upward transfer, the loss gradient satisfies*

$$\frac{\partial L}{\partial V_n} = \frac{\partial L}{\partial V_{p(n)}} R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}, \quad (3)$$

136 *with boundary condition $\partial L / \partial V_0 = \delta_0$. With post-voltage transfer $a_n = f_n(V_n)$, the local factor*
 137 *becomes $f'_n(V_n) R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}$. Because the graph is a tree, each compartment has a unique path to*
 138 *the soma. Defining the conductance-stage path gain*

$$\alpha_n^{\text{cond}} = \prod_{(i \rightarrow k) \in \text{path}(n \rightsquigarrow 0)} R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}, \quad \frac{\partial L}{\partial V_n} = \alpha_n^{\text{cond}} \delta_0, \quad (4)$$

139 *with $\alpha_0^{\text{cond}} = 1$ under identity transfer. In the post-voltage case, the exact effective path gain is*

$$\tilde{\alpha}_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightsquigarrow 0)} f'_i(V_i) R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}, \quad \frac{\partial L}{\partial V_n} = \tilde{\alpha}_n \delta_0, \quad (5)$$

140 *where $f'_i(V_i)$ is omitted for identity transfer.*

141 *Proof.* Apply the chain rule on the tree-structured computation graph using Prop. 1. □

142 The one-step recursion in Eq. (3) and the path product in Eq. (4) separate local conductance transfer
 143 from the non-local somatic error.

144 **Proposition 2** (Inhibitory control of conductance-stage path gain). *Let $\mathcal{A}(n)$ be the set of com-*
 145 *partments on the path from compartment n to the soma, excluding n and including the parent*
 146 *compartments through which the error must pass. Holding dendritic coupling conductances fixed, an*
 147 *added inhibitory conductance $\Delta G_k^I(x) \geq 0$ at any $k \in \mathcal{A}(n)$ changes the conductance-stage path*
 148 *gain by*

$$\frac{\alpha_n^{\text{cond}}(x; \Delta G^I)}{\alpha_n^{\text{cond}}(x; 0)} = \prod_{k \in \mathcal{A}(n)} \frac{g_k^{\text{tot}}(x)}{g_k^{\text{tot}}(x) + \Delta G_k^I(x)}, \quad \frac{\partial \log \alpha_n^{\text{cond}}}{\partial G_k^I} = \begin{cases} -R_k^{\text{tot}}, & k \in \mathcal{A}(n), \\ 0, & k \notin \mathcal{A}(n). \end{cases} \quad (6)$$

149 *Proof.* Substitute $R_k^{\text{tot}} = 1/g_k^{\text{tot}}$ into the path-gain product in Eq. (4). Adding inhibitory conductance
 150 changes only the denominator at compartments on the path from n to the soma, which gives the
 151 product ratio and the log derivative in Eq. (6). $\square$

152 Prop. 2 makes two directed predictions. First, inhibition is a path gate: it attenuates all credit paths
 153 below the inhibited compartment. Second, inhibition is useful for rank-1 local learning only when
 154 this attenuation makes the distribution of α_n^{cond} more uniform or better aligned with the task. If
 155 inhibition is too strong, too heterogeneous, or generated by a poorly calibrated inhibitory population,
 156 it can suppress useful signals and hurt learning. In the additive control the same inhibitory input
 157 changes voltages and local eligibilities, but it does not enter the conductance-stage path recursion as
 158 a multiplicative resistance gate.

159 **When does shunting compress path gains?** The attenuation identity above is not by itself a
 160 concentration result. Let $z_n = \log \alpha_n^{\text{cond}}$ be the log conductance-stage path gain for compartment n .
 161 If added inhibitory conductance contributes a path-dependent attenuation

$$\beta_n(x) = \sum_{k \in \mathcal{A}(n)} \log \left(1 + \frac{\Delta G_k^I(x)}{g_k^{\text{tot}}(x)} \right), \quad z'_n = z_n - \beta_n, \quad (7)$$

162 then, over compartments or examples,

$$\text{Var}(z') = \text{Var}(z) + \text{Var}(\beta) - 2 \text{Cov}(z, \beta). \quad (8)$$

163 Thus shunting narrows the log path-gain field exactly when

$$\text{Cov}(z, \beta) > \frac{1}{2} \text{Var}(\beta). \quad (9)$$

164 Eq. (7) follows by taking the logarithm of the product ratio in Prop. 2; Eq. (8) then applies the variance
 165 identity for $z - \beta$. Eq. (9) is an interpretive sufficient condition under the same averaging measure,
 166 not a primary empirical diagnostic. In finite trained networks, the covariance margin can disagree
 167 with CV and exact-error-rank summaries because these statistics average over different compartment
 168 and sample axes. We therefore use direct path-gain dispersion, exact-error compressibility, and
 169 inhibition-intervention measurements as the empirical tests, and report the covariance margin as a
 170 boundary check in the appendix.

171 **Corollary 1** (Local-Global Factorization). *The exact synaptic gradient at compartment n factorizes*
 172 *as:*

$$\frac{\partial L}{\partial g_i^{\text{syn}}} = \underbrace{x_i R_n^{\text{tot}} (E_i - V_n)}_{\text{synapse-local eligibility}} \cdot \underbrace{\frac{\partial L}{\partial V_n}}_{\text{compartment error}}, \quad (10)$$

173 *where the eligibility term depends only on quantities available at synapse i (presynaptic activity*
 174 *x_i , input resistance R_n^{tot} , driving force $E_i - V_n$), and the compartment error $\partial L / \partial V_n$ is the sole*
 175 *non-local quantity.*

176 The factorization in Eq. (10) isolates the only non-local quantity as $\partial L / \partial V_n = \alpha_n^{\text{cond}} \delta_0$ under identity
 177 upward transfer and $\partial L / \partial V_n = \tilde{\alpha}_n \delta_0$ when post-voltage transfer is enabled. The compartment error
 178 remains the only non-local term. Any practical local rule therefore depends entirely on how well
 179 a broadcast signal approximates that exact compartment error, which we evaluate in Sec. 4. In the
 180 models studied here, shunting improves this approximation by reducing and stabilizing R_n^{tot} , which
 181 narrows the spread of local sensitivities across the tree. In the empirical networks, each dendritic
 182 branch layer may optionally apply a post-voltage reactivation $f(V_n)$ before passing activity to the
 183 next stage; unless otherwise noted, the manuscript sweeps use a learnable bounded `param_tanh`
 184 reactivation.

3 Local Learning Rules

Broadcast error approximation. We approximate the exact compartment error $\partial L/\partial V_n$ (Corollary 1) with a broadcast signal e_n derived from the somatic/core error δ_0 . For a linear readout this reduces to $\delta_0 = W_{\text{dec}}^\top (\partial L/\partial \hat{y})$; for a nonlinear decoder we use the corresponding decoder-input Jacobian product $J_{\text{dec}}(h_{\text{core}})^\top (\partial L/\partial \hat{y})$ so that the broadcast remains defined in the same core-space coordinates. For rank-1 broadcast, $\bar{\delta}_b = d_0^{-1} \sum_{c=1}^{d_0} \delta_{0,bc}$ is computed separately for each example b and broadcast to all compartments of that example. We distinguish five feedback objects: **rank-1 shared broadcast**, $e_n = \bar{\delta} \mathbf{1}$; **neuron-wise broadcast**, $e_n = \delta_0$ when layer width matches the error dimension; **rank- K random broadcast**, $c = P_K \delta_0$ and $e_n = \sum_k q_{n,k} c_k$; **path-structured broadcast**, $e_n = \lambda_{\text{mix}} \bar{\delta} \mathbf{1} + (1 - \lambda_{\text{mix}}) \Gamma_n(\delta_0)$ with pathway roles motivated by vectorized dendritic teaching signals [44]; and the **transported oracle**, $e_n^{\text{transport}} = \tilde{\alpha}_n \delta_0 = \partial L/\partial V_n$, an analysis upper bound because it supplies the effective path gain from Theorem 1. Operationally, path-transport runs with reactivation enabled transport activation-space errors through parent reactivation slopes and multiply by the local $f'_n(V_n)$ before conductance updates; with identity transfer this reduces to $\alpha_n^{\text{cond}} \delta_0$. In the main feedforward architectures, hidden widths exceed the decoder-error dimension, so the default neuron-wise broadcast reduces to a rank-1 shared field (Appendix Table S10). Thus the main rank-1 result should be read as the architecture-induced low-bandwidth case, not as a claim that neuron-wise broadcasts were unavailable in principle. Under shunting, that deliberately low-bandwidth field is often sufficient on standard classification tasks. Routed tasks and rank-bridge controls in the appendix test the opposite regime, where rank-1 broadcast collapses branch identity and richer feedback is needed. A local-mismatch heuristic is included only as an appendix control because it performs much worse than the somatic-error broadcasts (Appendix A).

Three-factor learning rule (3F).

Definition 1 (3F Update). *For synaptic and dendritic conductances:*

$$\Delta g_j^{\text{syn}} = \eta \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n \rangle_B, \quad \Delta g_j^{\text{den}} = \eta \langle R_n^{\text{tot}} (V_j - V_n) e_n \rangle_B, \quad (11)$$

where $\langle \cdot \rangle_B$ denotes the batch average.

The three factors are presynaptic activity x_j (or a voltage difference), postsynaptic modulation through driving force and input resistance, and the broadcast error e_n . The same rule applies to excitatory and inhibitory synapses. The sign difference comes from the driving force $(E_j - V_n)$. We write Δg as the local gradient estimate supplied to the optimizer; the optimizer applies the usual descent step. Equivalently, one could absorb the minus sign into e_n and treat it as a descent signal.

Additive control. Rule (11) is the local gradient of the *shunting* voltage $V_n = \sum E_j g_j x_j / g_n^{\text{tot}}$. For the additive E/I control, inhibition enters as a fixed signed voltage contribution rather than through the conductance denominator:

$$V_n^{\text{add}} = \sum_{j \in E} g_j^E x_j^E - \sum_{j \in I} g_j^I x_j^I + \sum_k g_k^{\text{den}} V_k.$$

Equivalently, $\partial V_n^{\text{add}} / \partial g_j^{\text{syn}} = s_j x_j$ with $s_j = +1$ for excitatory and $s_j = -1$ for inhibitory synapses, and $\partial V_n^{\text{add}} / \partial g_k^{\text{den}} = V_k$. The additive local gradient therefore has no driving-force or R_n^{tot} terms ($R_n^{\text{tot}} \equiv 1$ by definition since there is no denominator). Throughout, each architecture uses the learning rule derived from its own forward-pass dynamics. This keeps the comparison architecture-matched rather than applying a shunting-derived rule to an additive model. In implementations with unconstrained raw parameters θ and $g = \text{softplus}(\theta)$, Eqs. (11)–(12) give conductance-space gradients; before assigning gradients to θ , we multiply by $\partial g / \partial \theta$. The exact-gradient reconstruction diagnostics include this parameterization factor.

Practical heuristic wrappers: 4F and 5F. 4F (branch-soma relevance proxy). In the conductance-stage recursion (4), the path gain α_n^{cond} attenuates the somatic error differently at each compartment. To compensate without explicitly computing that gain, 4F uses an online correlation proxy

$$r_n^{4F} = \text{Cov}(\bar{V}_n, \bar{V}_0) / (\sqrt{\text{Var}(\bar{V}_n) \text{Var}(\bar{V}_0)} + \varepsilon).$$

High r_n^{4F} up-weights branches whose voltages covary with the soma; low r_n^{4F} down-weights branches where the broadcast is likely to be a poor proxy. We keep 4F as a diagnostic intermediate. Empirically,

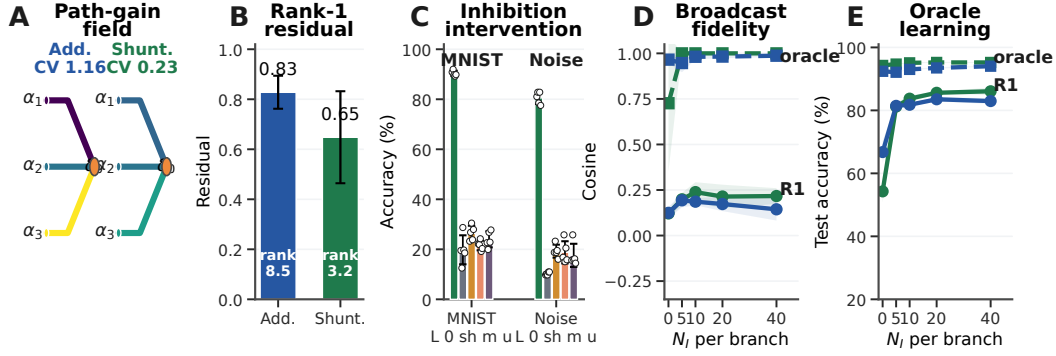


Figure 3: **Mechanistic chain from path gains to learning.** (A) Dendritic path-gain fields at matched inhibition. (B) Exact-error compressibility on MNIST: lower rank-1 residual and participation rank favor scalar broadcast. (C) Post-training inhibitory-conductance interventions (L: learned; 0/sh/m/u: zero, shuffled, mean-clamped, uniform matched). (D) Compartment-error fidelity on noise resilience; dashed lines show the transported-error oracle. (E) Transported-error learning recovers near-ceiling local learning where rank-1 broadcast fails. Error bars denote ± 1 s.d. across seeds.

231 this factor alone gives little improvement over 3F; the practical jump comes from the 5F confidence
232 factor below.

233 **5F (confidence-weighted 4F).** 5F adds a second heuristic factor

$$\phi_n = \text{Var}(V_n) / (\sigma_{\text{res}}^2 + \varepsilon),$$

234 where σ_{res}^2 is the residual variance of V_n after linear regression on the parent voltage. Operationally,
235 ϕ_n is a branch-wise confidence gate, clamped to $[0.25, 4.0]$ for stability. This factor is empirical, not
236 theorem-derived; its gain should be read as practical denoising rather than theoretical necessity. It is
237 positive and bounded, and rescales locally estimated reliability from compartment and parent-voltage
238 statistics rather than introducing a new error signal.

Definition 2 (5F Update).

$$\Delta g_j^{\text{syn}} = \eta r_n^{4F} \phi_n \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n \rangle_B, \quad \Delta g_j^{\text{den}} = \eta r_n^{4F} \phi_n \langle R_n^{\text{tot}} (V_j - V_n) e_n \rangle_B. \quad (12)$$

239 Eq. (12) is therefore the main practical LocalCA update, while the 3F rule is the theoretically derived
240 local gradient and 4F is the intermediate ablation. A feedback-alignment-style random-broadcast
241 argument is included in Appendix C; it is supportive but not central to the main mechanism claim.

242 4 Experiments

243 **Setup.** The experiments test five linked predictions: exact factorization should reconstruct autograd
244 gradients; shunting should improve LocalCA direction and scale relative to backpropagation; exact
245 compartment errors should be more compressible under shunting than under matched additive
246 controls; learned inhibition should carry sample-specific pathway structure rather than merely adding
247 conductance load; and replacing the broadcast with effective transported error should largely close
248 the local-learning gap. We evaluate capacity-calibrated supervised tasks and controlled sweeps that
249 isolate inhibition strength, rule family, broadcast mode, morphology, and architecture. The main
250 text uses MNIST [42], Fashion-MNIST [41], context gating, noise resilience, and a compressed
251 morphology-regime summary; CIFAR-10 [43] and routed-feedback diagnostics are reported in
252 the appendix (Figs. S7, S11). Main comparisons are architecture-matched additive vs. shunting
253 dendritic cores and local vs. backprop training; DFA is a supplementary baseline. Unless otherwise
254 stated, local runs use 5F with the default neuron-wise broadcast, which reduces to rank-1 shared
255 broadcast when hidden width exceeds the decoder-error dimension. All main-claim dendritic runs
256 use nonnegative conductance transforms and nonnegative first-layer synaptic drive; image inputs are
257 in $[0, 1]$, corrupted inputs are clipped back to $[0, 1]$, and context gating uses a nonnegative contextual
258 figure-ground MNIST construction. The transported-error condition is an oracle upper bound derived
259 from Theorem 1, not the proposed biological rule. Main mechanistic, oracle, and competence results
260 use 5 seeds per condition unless noted otherwise. For gradient diagnostics, weighted cosine is

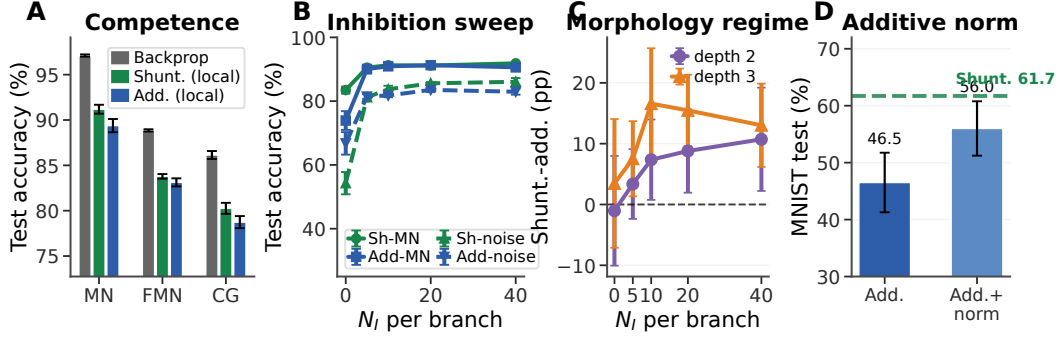


Figure 4: **Capacity-calibrated competence and regime dependence.** (A) Matched backprop reference performance vs. rank-1 5F LocalCA on MNIST, Fashion-MNIST, and context gating. (B) Inhibitory dose-response. (C) Morphology-dependent shunting advantage on noise resilience from a 3-seed sweep. (D) Separate stress-regime additive-normalization diagnostic: normalization improves additive LocalCA but remains below the shunting reference. Error bars denote ± 1 s.d. where available.

261 $\sum_p w_p \cos(g_p^{\text{local}}, g_p^{\text{bp}})$ with $w_p = \text{numel}(p) / \sum_q \text{numel}(q)$, and weighted scale mismatch uses the
 262 same weights on $|\log_{10}(\|g_p^{\text{local}}\| / \|g_p^{\text{bp}}\|)|$. Path-gain CV is the coefficient of variation of α_n^{cond}
 263 over the measured compartment field. Exact-error compressibility uses a matrix E whose rows are
 264 examples and whose columns are compartment–neuron coordinates; if σ_i are its singular values, the
 265 rank-1 residual is $\rho_1 = \sqrt{\sum_{i>1} \sigma_i^2} / \sqrt{\sum_i \sigma_i^2}$ and participation rank is $r_{\text{part}} = (\sum_i \sigma_i^2)^2 / \sum_i \sigma_i^4$.

266 From exact factorization to credit-signal fidelity.

267 To test whether these performance differences correspond to better credit signals, we compare
 268 local and backprop gradients on the same batch at the same parameter values. For each parameter
 269 tensor p , we compute directional alignment (cosine similarity) and scale mismatch, defined as
 270 $|\log_{10}(\|g_p^{\text{local}}\| / \|g_p^{\text{bp}}\|)|$, aggregated by parameter count.

271 Across the MNIST gradient-dynamics diagnostic, shunting is more aligned with backprop and closer
 272 in norm than the additive control at the final checkpoint: weighted cosine rises from 0.005 ± 0.026
 273 to 0.100 ± 0.033 , and scale mismatch falls from 0.50 ± 0.05 to 0.26 ± 0.05 . Figure 2A confirms that
 274 the implementation matches the theory under the nonnegative-input setup: reconstructing gradients
 275 from Corollary 1 using exact compartment errors yields weighted relative reconstruction error below
 276 2×10^{-7} and weighted scale mismatch below 4×10^{-8} across runs. In a few low-norm parameter
 277 blocks, cosine becomes numerically unstable even though the reconstruction itself is essentially
 278 exact, so we report the norm-based diagnostics in the figure. Figure 2D shows the same gradient-
 279 alignment comparison over training: additive alignment remains near zero at the final checkpoint
 280 rather than failing only at one static snapshot. Appendix Fig. S3 verifies that the corresponding local
 281 and backprop gradient norms remain finite throughout learning, ruling out a trivial zero-gradient
 282 explanation. At the level of the theoretically derived 3F rule, the central non-local approximation is
 283 therefore the broadcast substitute for $\partial L / \partial V_n$.

284 **Mechanistic chain: path gains, broadcast fidelity, and oracle transport.** Figure 3 provides the
 285 central empirical chain. At matched inhibition ($N_I=5$), shunting narrows the MNIST conductance-
 286 stage path-gain field (CV 0.23 vs. 1.16 for additive) and makes the exact compartment-error matrix
 287 more compressible: the best rank-1 residual drops from 0.83 ± 0.07 to 0.65 ± 0.18 , while participation
 288 rank drops from 8.5 ± 3.3 to 3.2 ± 1.2 . This is the diagnostic matched to the paper’s main claim:
 289 rank-1 broadcast is useful when the exact error field is closer to low rank.

290 The inhibition intervention shows that the learned inhibitory field is necessary for the trained compu-
 291 tation and is not interchangeable with a matched conductance load: zeroing, shuffling, or clamping
 292 learned branch inhibition drops MNIST from about 90% to 20–26% and noise resilience from about
 293 81% to 10–19%. The fidelity and oracle panels probe the same broadcast bottleneck: rank-1 broadcast
 294 tracks the exact compartment error better under shunting once inhibition is present, transported error
 295 reaches about 0.95–1.0 fidelity, and transported learning lifts both the shunting and additive networks

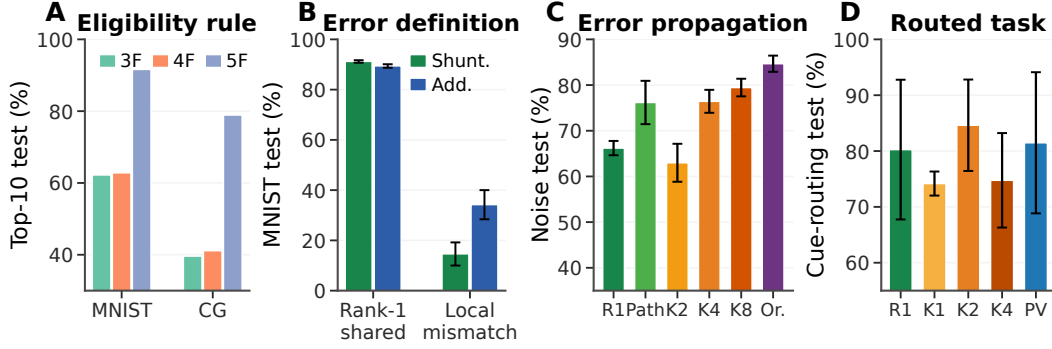


Figure 5: **Rule and feedback controls.** (A) Eligibility-rule ablation: 3F is the theorem-derived local eligibility, 4F adds branch–soma relevance, and 5F adds the confidence wrapper. (B) Error-source ablation: rank-1 shared somatic error works, whereas a local-mismatch surrogate collapses learning. (C) Feedback-bandwidth ladder on noise resilience: R1 is rank-1, Path is recursive path-propagation, K2/K4/K8 are random low-rank, and Or. is exact effective transported error. (D) Cue routing tests branch-specific credit: R1 is rank-1, K1/K2/K4 are random low-rank, and PV is pathway-vector feedback; higher rank helps but PV does not yet beat the best unstructured low-rank control.

296 to 92–95% on noise resilience. CIFAR-10 and direct-I versus explicit-I path-gain probes remain
 297 appendix diagnostics (Fig. S7; Table S4).

298 **Capacity-calibrated competence.** In the capacity-calibrated regime, shunting LocalCA remains
 299 within around 5 percentage points of matched backpropagation on MNIST, Fashion-MNIST, and
 300 context gating (Table S2; Fig. 4). In a separate lower-capacity stress-regime diagnostic, Fig. 4D shows
 301 that additive normalization improves additive LocalCA but remains below the shunting reference,
 302 so the gain is not explained by generic voltage normalization alone. Figure 5 summarizes the rule
 303 and feedback controls: 5F is the strongest practical optimizer, somatic rank-1 error is essential, and
 304 higher feedback bandwidth improves the noise-resilience diagnostic.

305 **Regime dependence of the shunting advantage.** With matched rank-1 broadcast, shunting is
 306 modestly better on MNIST, Fashion-MNIST, and context gating, including 0.803 ± 0.006 vs. $0.787 \pm$
 307 0.007 for additive on context gating. The gap grows on inhibition-sensitive noise resilience once
 308 inhibitory conductance is present (Fig. 4B). Morphology shifts the useful inhibitory operating regime:
 309 different tree depths and branch factors peak at different N_I , and the shunting advantage is not
 310 monotonic in inhibition (Fig. 4C). Appendix Fig. S5 extends this boundary check to depth scaling
 311 and noisy error broadcasts.

312 5 Discussion and Limitations

313 We studied local credit assignment in conductance-based dendritic networks with explicit E/I con-
 314 ductance synapses, shunting inhibition, and tree-structured branching. Exact gradients separate
 315 synapse-local eligibility–presynaptic activity, driving force, and input resistance—from one path-
 316 specific compartment error, making LocalCA an error-approximation problem.

317 The experiments support a coherent mechanism. Branching creates credit paths, E/I synapse banks
 318 determine the conductance state in each compartment, and shunting modulates input resistance along
 319 those paths. In the regimes studied here, this makes exact errors more rank-1 compressible, improves
 320 LocalCA to backprop alignment, and reduces scale mismatch. Transported-error oracles largely close
 321 the local-learning gap, while inhibition interventions show that learned inhibitory conductance carries
 322 sample-specific pathway structure rather than only global conductance scale. Appendix Table S5
 323 states direct biological predictions of the same path-gain mechanism.

324 The limits are important. Shunting is not universally beneficial, and more inhibition is not always
 325 better. The practical 5F rule is empirical; Eq. (9) is interpretive; activation shape, morphology,
 326 task structure, and broadcast rank all change the operating regime. We do not claim competitive
 327 CIFAR-10 performance, a recurrent inhibitory circuit model, or that scalar broadcast suffices for
 328 routed/pathway-specific tasks.

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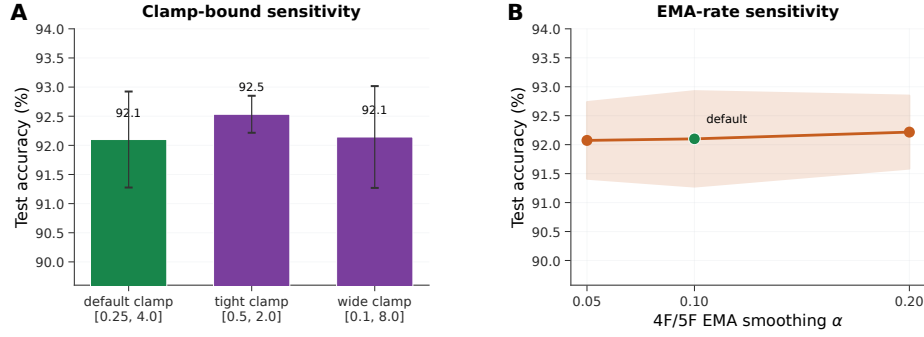


Figure S1: **5F stabilizer sensitivity.** (A) Changing the confidence-factor clamp from the default [0.25, 4.0] to tighter [0.5, 2.0] or wider [0.1, 8.0] bounds leaves MNIST test accuracy near 92%. (B) Changing the 4F/5F EMA smoothing rate over $\alpha \in \{0.05, 0.10, 0.20\}$ gives similarly stable performance. Error bars and bands are ± 1 s.d. across 3 seeds.

A Supplementary Results

Appendix overview. The appendix is organized by how directly each result supports the main mechanism. It first reports rule, feedback, gradient, competence, and baseline controls; then gives harder-data, morphology, inhibition, and routed-feedback diagnostics; and finally provides implementation, theory, and task details needed for reproducibility.

Boundary diagnostics. Several controls are intentionally negative or mixed. The log-gain covariance margin from Eq. (9) is not used as a positive main result, and the pathway-vector feedback variant does not beat the best unstructured low-rank cue-routing control. These diagnostics delimit the claim rather than smoothing over failure modes.

Rule-Family Ranking

This table keeps the 3F/4F/5F optimizer distinction explicit: the theorem-derived eligibility is necessary for the story, but the reported LocalCA competence results use the empirical 5F stabilizer.

Dataset	Rule	Top-10 sweep valid	Top-10 sweep test
MNIST	3F	0.611	0.622
MNIST	4F	0.620	0.628
MNIST	5F	0.912	0.916
Context gating	3F	0.398	0.396
Context gating	4F	0.411	0.411
Context gating	5F	0.807	0.789

Table S1: **Rule-family ranking.** Mean over the ten best validation-selected configurations from local-competence sweeps; this is not top-10 classification accuracy.

5F Stabilizer Sensitivity

Because the headline LocalCA runs use the empirical 5F stabilizer, we also tested whether performance depends sharply on the confidence clamp or the EMA rate used to estimate branch statistics. Across three-seed MNIST sensitivity checks, tightening or widening the clamp and varying the EMA rate leaves test accuracy near the default setting. This does not make 5F theorem-derived; it supports interpreting it as a bounded local reliability preconditioner rather than a brittle tuned error source.

Dataset	BP ceiling	Shunting local	Additive local	BP gap
MNIST	0.971	0.912 ± 0.005	0.894 ± 0.007	5.9%
Fashion-MNIST	0.889	0.838 ± 0.003	0.831 ± 0.004	5.1%
Context gating	0.861	0.803 ± 0.006	0.787 ± 0.007	5.8%

Table S2: **Local competence.** Backprop ceilings are matched-capacity shunting references from dedicated 5-seed standard-training refreshes for MNIST and context gating and from the dedicated 5-seed Fashion-MNIST competence sweep. Local values are 5F with rank-1 broadcast and local decoder updates. Context gating additionally uses an HSIC auxiliary objective (weight 0.01; Appendix C). Error bars on local values: ± 1 s.d. across 5 seeds.

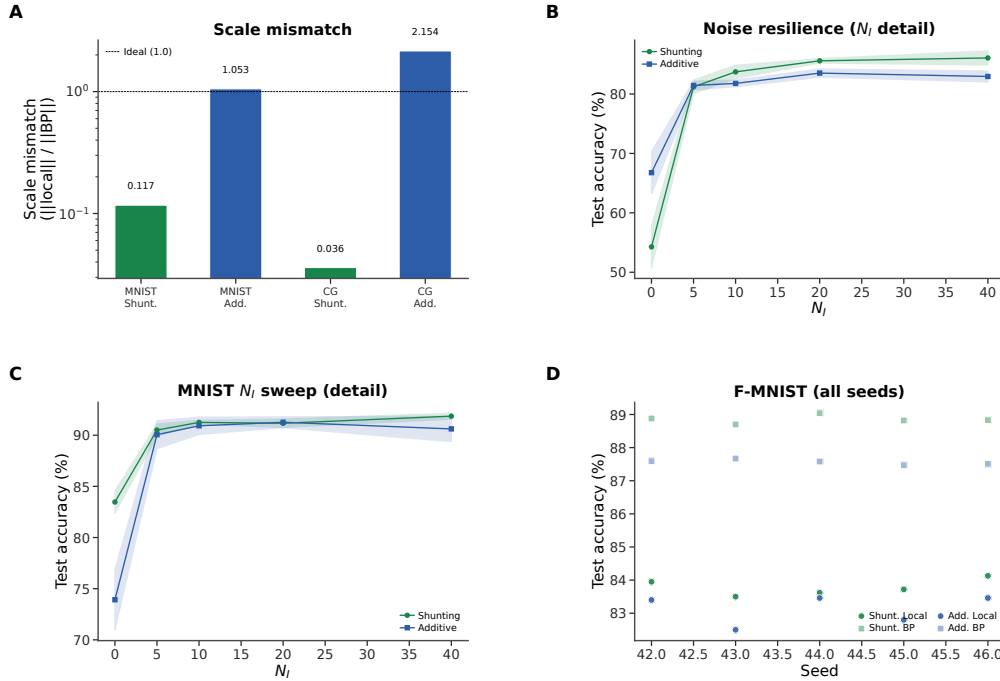


Figure S2: **Extended gradient and N_I analysis (supplementary).** (A) Scale mismatch bars from a separate matched-weight static diagnostic; these values should not be compared directly to the checkpointed final-state diagnostic in Fig. 2. Shunting achieves near-ideal scale (0.117); additive exhibits order-of-magnitude distortion (> 1.0). (B) Noise-resilience N_I dose-response with error bands (± 1 s.d.). (C) MNIST N_I dose-response detail with error bands. (D) Fashion-MNIST individual seeds for all conditions, showing consistency across runs.

449 Local Competence and Regime Dependence

450 Extended Gradient Analysis and N_I Sweep Detail

451 **Alignment dynamics are not a zero-gradient artifact.** To check whether low additive alignment
452 could be caused by vanishing gradients, we analyzed a separate 3-seed checkpoint trajectory with
453 parameter-level LocalCA and backprop gradients. The additive local gradients are nonzero throughout
454 training (weighted norm range 9.5×10^{-4} – 9.5×10^{-3}), and the corresponding backprop gradients are
455 also nonzero (2.3×10^{-5} – 5.7×10^{-3}). The low additive cosine therefore reflects directional and scale
456 mismatch, not absent gradients: additive 5F cosine stays near zero from epoch 0 to epoch 50 (0.018
457 to 0.005), while its local/backprop norm ratio falls from about 8.8×10^2 to 0.69.

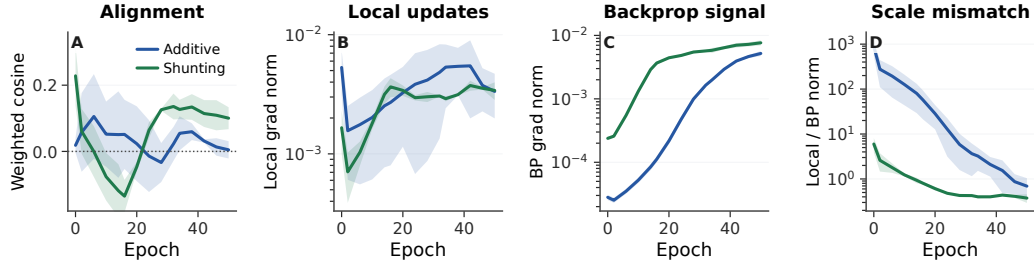


Figure S3: **Alignment dynamics with explicit gradient-norm checks.** (A) Weighted LocalCA to backprop cosine over training for the 5F rule. (B,C) Weighted local and backprop gradient norms stay nonzero for both additive and shunting models. (D) The additive control begins with a much larger scale mismatch, then approaches the backprop scale without recovering strong directional alignment. Shading is s.d. across 3 seeds.

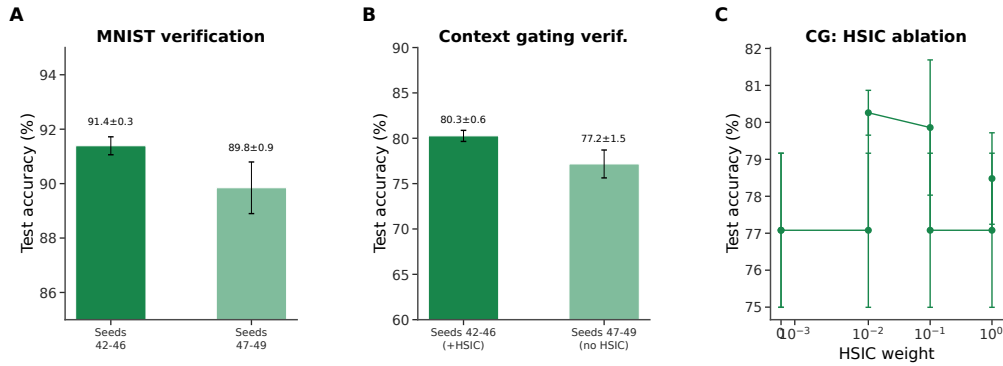


Figure S4: **Verification and reproducibility (supplementary).** (A) MNIST verification: main seeds (42–46) yield $91.4 \pm 0.3\%$; held-out seeds (47–49) yield $89.8 \pm 0.9\%$, confirming generalization. (B) Context gating verification: main seeds (+HSIC) $80.3 \pm 0.6\%$; held-out seeds (no HSIC) $77.2 \pm 1.5\%$. The roughly 3 pp gap reflects HSIC removal, not seed sensitivity. (C) HSIC weight ablation on context gating: moderate weights (0.01–0.1) perform best.

458 Verification and Reproducibility

459 Additional Stress Tests

460 FA/DFA Baseline Comparison

461 CIFAR-10 Results

462 Table S3 provides the exact values for the appendix CIFAR-10 figure. CIFAR-10 is used only as
 463 a harder-data mechanistic diagnostic: under flattened inputs, nonnegative conductance constraints,
 464 and a compact passive dendritic core, the matched backprop ceiling is the relevant reference rather
 465 than modern vision accuracy. In this stronger compact direct-I-stream family, shunting performs well
 466 under standard training (49.5% vs. 48.3% for additive), so the harder-data stress test does not unfairly
 467 penalize the shunting architecture. The broadcast-fidelity ladder also survives once decoder-aware
 468 soma mapping is used: additive LocalCA reaches 25.5% under rank-1 per-soma broadcast and
 469 32.4% under transported error, while shunting rises from 29.7% under rank-1 broadcast to 35.0%
 470 with random low-rank $K=4$ and 47.2% with transported error, nearly matching the 49.5% shunting
 471 backprop ceiling. Removing learned I-to-E conductance lowers the shunting ceiling by 5.7 pp and
 472 transported LocalCA by 8.6 pp, making this a harder-data diagnostic for inhibitory conductance as
 473 part of the useful credit geometry rather than just an added architectural detail.

474 Additive Normalization Control

475 This control separates generic additive voltage normalization from shunting conductance dynamics in
 476 a lower-capacity MNIST stress regime.

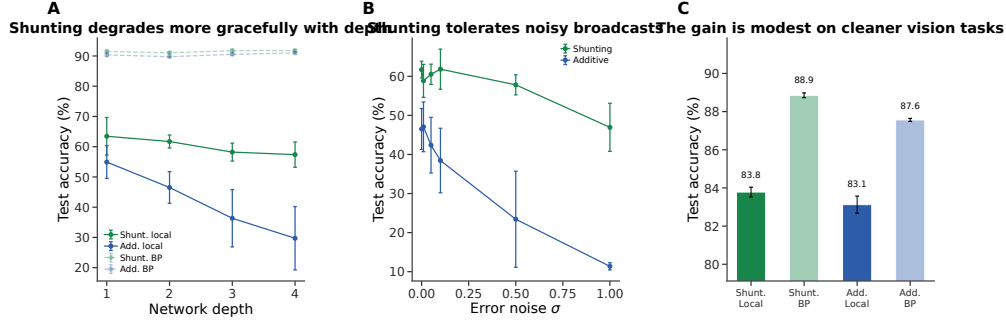


Figure S5: **Additional stress tests.** (A) Depth scaling: shunting local (green) degrades more gracefully from 63.5% to 57.4%; additive local (blue) drops from 54.9% to 29.7%. Backprop ceilings (dashed) remain stable. (B) Broadcast-noise robustness: shunting remains stable for $\sigma \leq 0.1$; additive degrades rapidly and reaches chance at $\sigma = 1$. (C) Fashion-MNIST: the shunting gain is modest on this cleaner benchmark (83.8% local vs. 88.9% backprop), consistent with the regime-dependent story in the main text.

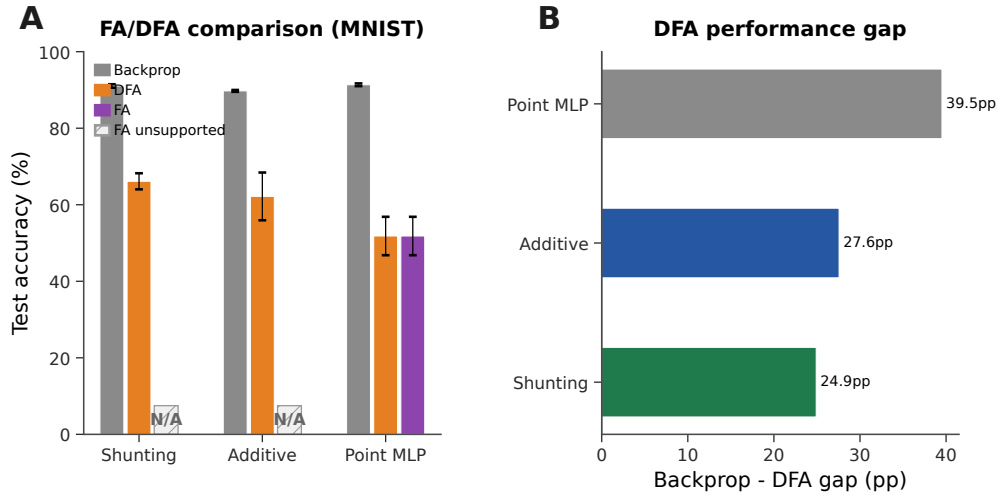


Figure S6: **Feedback alignment baselines (MNIST).** (A) Grouped comparison: standard backprop (neutral gray), DFA (orange), and FA (purple). Hatched N/A markers indicate that FA is not implemented for the block-structured dendritic architectures because the random feedback matrices are not dimensionally compatible with the tree-structured branch updates. DFA achieves 66.1% on shunting vs. 62.2% additive vs. 51.8% point MLP. (B) Backprop–DFA gap: dendritic architectures (24.9–27.6 pp) show smaller gaps than point MLPs (39.5 pp), suggesting conductance-based architecture is partially compatible with random feedback.

477 **Morphology $\times$ Inhibition Regime Map**

478 This regime map tests whether tree geometry changes the inhibitory operating point at which shunting
 479 helps. It supports the claim that morphology changes the useful inhibition regime; a fuller mechanistic
 480 account would require matched theory diagnostics for every cell in the grid.

481 **Input-Mode Probe: Direct Inhibitory Stream vs. Explicit Inhibitory Cells**

482 The main experiments use input-driven inhibitory streams: the transfer layer provides nonnegative
 483 excitatory and inhibitory channels from the external input, and excitatory dendrites receive learned
 484 I-to-E conductance banks directly. To test whether the same mechanism can be generated by an
 485 explicit feedforward inhibitory population, we ran a one-feedforward-layer noise-resilience probe
 486 that preserves the relevant dendritic depth ([3, 3] morphology). In this setting, explicit inhibitory cells
 487 reach $94.2 \pm 0.2\%$ under path-transport LocalCA when their dendrites use the same update policy as
 488 excitatory dendrites, $93.8 \pm 0.2\%$ when those inhibitory-cell updates are frozen, and $95.1 \pm 0.3\%$

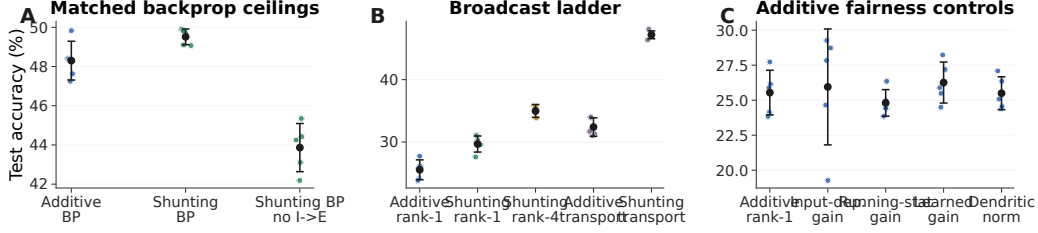


Figure S7: **CIFAR-10 control ladder in the compact direct-I-stream family.** (A) Matched backprop ceilings. With nonnegative inputs and a nonlinear decoder, shunting slightly exceeds additive under standard training, and removing learned I-to-E conductance lowers the shunting ceiling. (B) Broadcast ladder. Rank-1 broadcast remains weak on this harder dataset, rank-4 improves partially, and exact effective transported error nearly reaches the shunting backprop ceiling. (C) Additive fairness controls. Input-dependent gain, running-stat gain, learned gain, and dendritic normalization do not rescue additive rank-1 broadcast. This figure is a harder-data diagnostic, not a claim of competitive CIFAR-10 benchmarking.

Condition	Core	Training / broadcast	I-to-E	Test acc.
standard	additive	backprop	yes	48.3 $\pm$ 1.0
standard	shunting	backprop	yes	49.5 $\pm$ 0.4
standard, no learned inhibition	shunting	backprop	no	43.9 $\pm$ 1.2
rank-1	additive	5F per-soma	yes	25.5 $\pm$ 1.6
rank-1, input-dependent gain	additive	5F per-soma	yes	25.9 $\pm$ 4.1
rank-1, running-stat gain	additive	5F per-soma	yes	24.8 $\pm$ 0.9
rank-1, learned gain	additive	5F per-soma	yes	26.3 $\pm$ 1.5
rank-1, dendritic normalization	additive	5F per-soma	yes	25.5 $\pm$ 1.2
rank-1	shunting	5F per-soma	yes	29.7 $\pm$ 1.3
rank-1, no learned inhibition	shunting	5F per-soma	no	22.3 $\pm$ 0.7
transported error	additive	5F path transport	yes	32.4 $\pm$ 1.5
rank-4	shunting	5F low-rank	yes	35.0 $\pm$ 1.0
transported error	shunting	5F path transport	yes	47.2 $\pm$ 0.6
transported error, no learned inhibition	shunting	5F path transport	no	38.6 $\pm$ 1.0

Table S3: **Exact CIFAR-10 values for the 70-run control ladder (5 seeds per condition, validation-best epoch).** All rows use flattened CIFAR-10 in $[0, 1]$, nonnegative transfer outputs, positive conductance transforms, a compact depth-4 dendritic core, and a nonlinear decoder. LocalCA maps output error back to decoder-input/soma coordinates through the decoder Jacobian. The useful conclusions are narrow but important: learned shunting slightly improves the matched backprop ceiling, learned I-to-E conductance is necessary for the strongest shunting results, transported error nearly closes the shunting LocalCA gap, and additive rank-1 fairness controls do not explain the transported-shunting result.

under standard backprop. Direct input-driven inhibition gives the expected broadcast-fidelity ladder on the same one-layer architecture: rank-1 broadcast reaches $85.2 \pm 0.7\%$, while exact effective path transport reaches $95.2 \pm 0.0\%$. Thus explicit inhibitory cells can generate useful path-gain structure once the architecture isolates the dendritic-path question.

Post-training inhibition intervention and exact-error rank. We next tested whether learned inhibitory conductance carries task-specific structure rather than only changing global scale. We replayed trained shunting LocalCA checkpoints while replacing the branch-level inhibitory conductance with four post-training interventions: zero inhibition, sample-shuffled inhibition, per-branch batch means, or one uniform matched mean. On a 512-example held-out subset, learned MNIST inhibition at $N_I=5$ gives $90.4 \pm 1.1\%$ accuracy, while the four interventions drop to 19.8–26.4%. On noise resilience at $N_I=10$, learned inhibition gives $80.5 \pm 2.4\%$, while the interventions drop to 10.3–19.3%. Thus the inhibitory field is not interchangeable with a matched global conductance load. We also computed the SVD of exact compartment-error matrices across samples and compartments. At matched MNIST $N_I=5$, shunting has lower rank-1 residual than additive (0.65 ± 0.18 vs. 0.83 ± 0.07) and lower participation rank (3.2 ± 1.2 vs. 8.5 ± 3.3), supporting the path-gain compressibility interpretation.

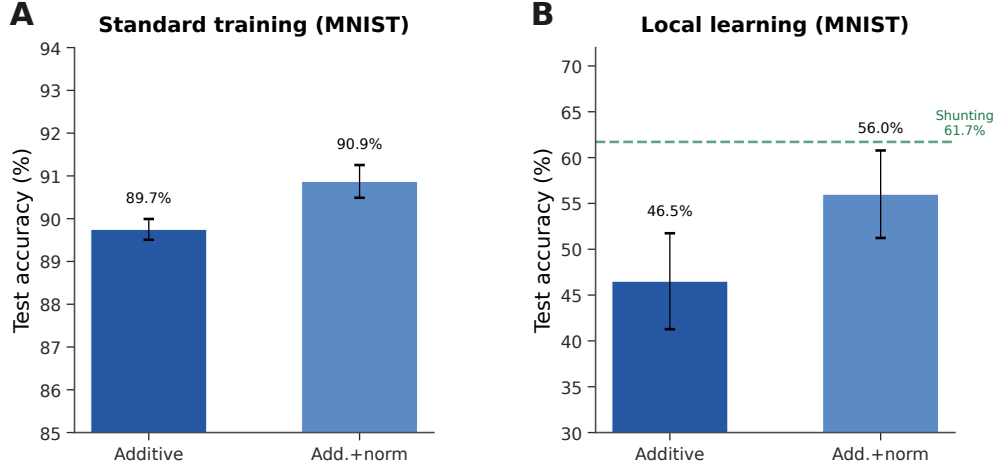


Figure S8: **Additive + normalization control (MNIST stress-regime diagnostic).** (A) Standard backprop: normalization provides a small boost (89.7% $\rightarrow$ 90.9%). (B) Local learning: normalization improves additive from 46.5% to 56.0% (+9.5 pp), partially closing the gap to shunting (61.7%, dashed green). These lower local accuracies come from a separate stress-regime diagnostic, not from the capacity-calibrated competence setting in Table S2.

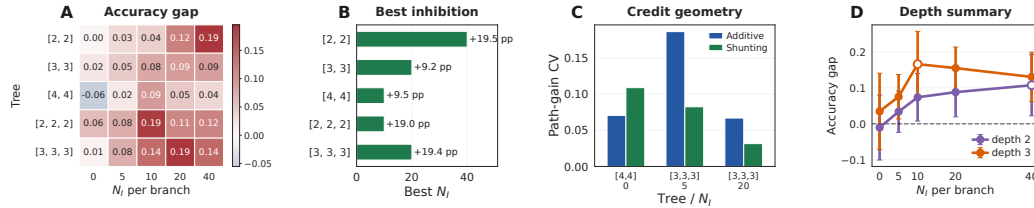


Figure S9: **Morphology shifts the useful inhibitory regime.** (A) Shunting-minus-additive LocalCA accuracy gap across dendritic morphologies and inhibitory synapse counts. (B) Best inhibitory count for each morphology. (C) Representative path-gain diagnostic for selected morphology cells, used to connect the performance map to credit geometry. (D) Average shunting advantage by depth, with markers showing the peak inhibitory count for each depth.

We also computed the direct log-gain covariance margin in Eq. (9) on the selected morphology diagnostic checkpoints. The margin is zero when no inhibitory conductance is present and slightly negative in the inhibited shunting selections (-2.5×10^{-3} , -5.0×10^{-4} , and -2.8×10^{-4} at $N_I=5, 20, 40$). We therefore do not use this covariance condition as a positive main result. Its role is to clarify when shunting should compress gains; the empirical claim rests on the measured comparative path-gain dispersion, exact-error rank, and inhibition-intervention diagnostics.

Experimental Predictions

The model is not a detailed cell-type circuit model, but its path-gain identity gives falsifiable dendritic predictions. These predictions are naturally aligned with branch-level dendritic computation and compartment-targeted inhibition in pyramidal neurons [5–7, 12].

Cue-Routing Feedback-Rank Diagnostic

Cue routing tests the regime where one shared teaching signal is expected to fail: context determines which noisy cue is reliable, so different pathways should receive different credit. Figure S11 shows that the benchmark is learnable and that moving beyond rank-1 feedback helps, but the path-structured variant does not yet beat the best unstructured low-rank setting. We therefore interpret this result as a failure-mode and rank-requirement diagnostic, not as a solved structured-feedback method.

Inhibitory input mode	Training / broadcast	Test accuracy
Direct inhibitory stream	LocalCA, per-soma rank-1	0.852 ± 0.007
Direct inhibitory stream	LocalCA, exact effective path transport	0.952 ± 0.000
Explicit I cells	LocalCA, path transport, train I-cell dendrites	0.942 ± 0.002
Explicit I cells	LocalCA, path transport, freeze I-cell dendrites	0.938 ± 0.002
Explicit I cells	standard backup	0.951 ± 0.003

Table S4: **One-feedforward-layer input-mode path-gain probe on noise resilience (3 seeds).** All rows use one excitatory dendritic population with $[3, 3]$ morphology and nonnegative inputs. The direct-stream rows have no explicit inhibitory neurons; the input stream drives learned branch-level inhibitory conductance banks directly. The explicit-I rows use a matched inhibitory population with $[3, 3]$ morphology. Exact effective path transport closes the direct-stream rank-1 gap, and explicit inhibitory cells nearly match both direct-stream path transport and the explicit-I backup ceiling.

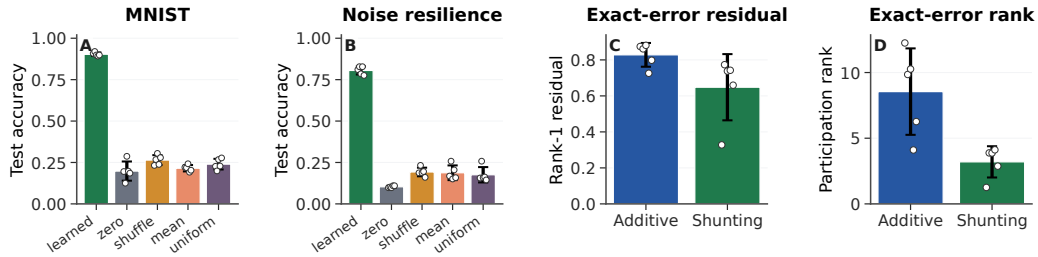


Figure S10: **Post-training inhibition intervention and exact-error compressibility.** (A,B) Post-training inhibitory-conductance interventions on trained shunting LocalCA checkpoints. Accuracy is measured on a 512-example held-out subset; dots show seeds. Removing, shuffling, or clamping learned inhibition collapses performance, showing that the learned inhibitory field carries sample-specific task structure. (C,D) SVD diagnostics of the exact compartment-error matrix on matched MNIST $N_I=5$ runs. Shunting has a smaller rank-1 residual and lower participation rank than additive, so the exact error field is more compressible. Error bars are s.d. across 5 seeds.

Random Low-Rank Broadcast Channels

These controls test whether failures of rank-1 broadcast reflect insufficient feedback bandwidth rather than a failure of local eligibility itself.

B Implementation Details

Biological Plausibility Assumptions

The theorems and rules above rest on a small set of explicit assumptions. We state them as a checklist so that readers can see exactly what is assumed and where each assumption is used:

A1 (Steady-state voltage). We use the steady-state solution of the passive cable equation (Eq. (1)). Temporal dynamics are not modeled; all gradients, path gains, and fidelity measurements refer to the conductance-stage voltage at steady state.

A2 (Tree topology). Each dendritic cell is a rooted tree with the soma as the root. There are no recurrent connections within the dendrite, which makes the path from any compartment to the soma unique and is what allows the closed-form path gains α_n^{cond} and $\tilde{\alpha}_n$ in Theorem 1.

A3 (Nonnegative conductances). Synaptic and dendritic conductance parameters are pushed through a positive transform (`softplus`), so $g_j^{\text{syn}}, g_j^{\text{den}} \geq 0$ at every optimisation step. This is a biophysical constraint and is also what keeps the shunting denominator bounded away from zero.

A4 (Nonnegative presynaptic drive). The first synaptic drive x is nonnegative, implemented either by nonnegative inputs or an explicit ReLU transfer stage. Together with A3 this enforces the positive-input condition used for all main sweeps.

Model implication	Biological prediction
Path-local inhibitory conductance changes α_n^{cond} only for descendants whose credit path crosses the inhibited compartment. The log derivative $\partial \log \alpha_n^{\text{cond}} / \partial G_k^I = -R_k^{\text{tot}}$ predicts stronger credit gating in high-resistance compartments. Rank-1 broadcast succeeds when the exact compartment-error matrix is low-rank and fails when branch identity matters.	Branch-local inhibitory perturbation should distort plasticity or teaching-signal efficacy for synapses below the perturbed compartment more than for sister branches. The same inhibitory conductance change should have larger effects on low-conductance/high-resistance branches than on already high-conductance branches. Tasks requiring route- or context-specific branch credit should need richer dendritic teaching signals than a single scalar modulatory broadcast.

Table S5: **Experimental predictions from the path-gain mechanism.** These are qualitative biological predictions, not claims tested directly in this paper.

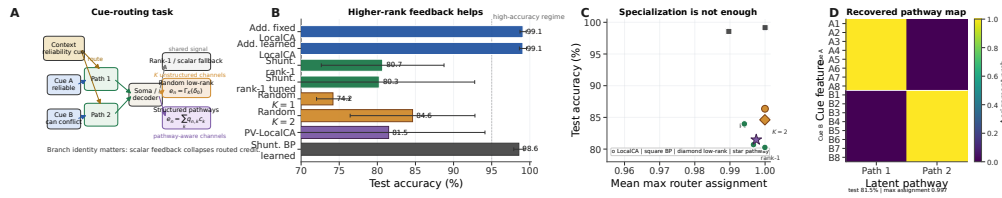


Figure S11: **Structured feedback when rank-1 broadcast fails.** (A) Cue-routing task schematic. Context determines which noisy cue is reliable on each trial, so different dendritic pathways should receive different teaching signals. (B) Five-seed comparison of rank-1, random low-rank, and pathway-structured feedback. Moving beyond rank-1 helps, while the best structured repair remains open. (C) Test accuracy vs. router specialization across learned-router conditions. High specialization alone does not guarantee success. (D) Recovered pathway assignments for a representative learned-router run. Cue-A and cue-B features segregate into separate latent pathways even when feedback quality, rather than router discreteness, is the limiting factor.

A5 (Pre-reactivation upward transfer). Theorem 1 is stated for the pre-reactivation compartment voltage. A post-voltage reactivation f (e.g. `param_tanh`) is handled by replacing α_n^{cond} with the effective gain $\tilde{\alpha}_n$ that includes $f'(V)$ factors along the path. The compartment error remains the sole non-local quantity; only the effective path gain acquires local activation-derivative factors.

A6 (Single perisomatic broadcast). Every synapse receives one non-local scalar or low-rank vector signal delivered from the soma; all other ingredients of the update rule (eligibility traces, driving force, input resistance, covariance modulators) are computable from quantities available at the synapse or compartment.

Each synapse therefore has access to: presynaptic activity x_j (local), compartment voltage V_n (local membrane potential), reversal potential E_j (fixed by receptor type), and total input resistance $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$ (computable from local conductances). The 4F modulator r_n^{4F} requires online estimation of voltage covariance (biologically plausible via slow calcium signals); the 5F factor ϕ_n requires a linear regression proxy (implementable via eligibility traces). The only non-local quantity is the broadcast error e_n , which requires a top-down or neuromodulatory signal from the soma to dendritic compartments. We assume per-neuron resolution (one error per output neuron) for the per-soma broadcast mode.

Dendritic Structure

We use the morphology notation $[b_1, b_2, \dots, b_D]$ for rooted dendritic trees, where each factor gives the fan-in from one dendritic stage to the next on the path to the soma. For example, $[3, 3]$ means that each soma aggregates 3 proximal branches and each proximal branch aggregates 3 distal branches, for 9 distal leaves per soma. In the main feedforward models, synaptic inputs terminate on dendritic branches rather than directly on the soma. The main setting uses direct input-driven inhibition: the transfer layer creates nonnegative excitatory and inhibitory streams from the same external input, and

Setting	Broadcast	Test accuracy
<i>Noise resilience, shunting, rank bridge</i>		
per-soma	rank-1 per-soma	0.662 ± 0.016
path propagation	recursive attenuation proxy	0.762 ± 0.047
low-rank	$K=1$	0.453 ± 0.126
low-rank	$K=2$	0.630 ± 0.042
low-rank	$K=4$	0.764 ± 0.025
low-rank	$K=8$	0.795 ± 0.019
path transport	exact effective path transport	0.847 ± 0.018
<i>Cue routing, learned shunting router, rank/structure sweep</i>		
per-soma	tuned rank-1	0.803 ± 0.125
low-rank	$K=1$	0.742 ± 0.022
low-rank	$K=2$	0.846 ± 0.082
low-rank	$K=4$	0.748 ± 0.085
pathway-vector	tuned structured rank-2	0.815 ± 0.126

Table S6: **Rank bridge and rank-structure comparison.** On shunting noise resilience, both recursive path propagation and higher-rank random low-rank broadcast close a substantial fraction of the gap between rank-1 per-soma and exact effective path transport, with low-rank continuing to improve up to $K=8$. On cue routing, the learned-router sweep shows that moving beyond rank-1 helps, but the best unstructured low-rank setting ($K=2$) is still slightly better than the tuned structured pathway-vector variant. We therefore interpret routed credit assignment as an open higher-rank problem rather than as a solved pathway-vector rescue.

learned inhibitory conductance banks attach directly to excitatory dendritic branches. These models therefore have branch-level inhibitory conductance but no separate inhibitory neuron population. The explicit-I probe instead instantiates feedforward inhibitory cells, which transform excitatory activity before projecting inhibitory conductance back onto excitatory dendrites. In that setting the inhibitory cells are also dendritic modules and their synaptic conductances are trained by the same local rule family. The additive and shunting cores are architecture-matched at the tree level; they differ in the branch-level voltage integration rule rather than in whether a dendritic tree exists. Matched comparisons use the same tree, synapse counts, nonnegative parameterization, transfer stage, optimizer protocol, and seed sets. In all reported dendritic runs, synaptic and dendritic conductance parameters are pushed through positive transforms, so conductances remain nonnegative; unconstrained decoder weights are separate readout parameters and are not interpreted as conductances.

Post-Voltage Reactivation

The conductance stage produces a pre-reactivation compartment voltage V_n , after which each branch layer may apply a monotone scalar nonlinearity. The main sweeps use a learnable bounded firing-rate transform, $(\tanh(m(V-b))+1)/2$, together with a nonnegative transfer stage so that the first synaptic drive remains nonnegative even after input corruption or structured task noise. Path-transport oracle runs with reactivation enabled use the effective transported error: parent-to-child transport includes the parent activation derivative, and the local update multiplies by the child derivative before applying conductance-space eligibility. Most main feedforward sweeps use a linear decoder, while the CIFAR extension uses a nonlinear decoder and therefore maps output error back to decoder-input/soma space through the decoder Jacobian before applying LocalCA. Unless explicitly overridden, additive and shunting cores can start from different reactivation initializations; activation-fairness ablations therefore match additive initialization to the shunting setting when comparing alternative reactivation choices directly. The appendix activation audit additionally fixes the bounded reactivation after initialization to isolate activation shape from activation plasticity.

Units and Parameterization

Decoder Update Modes

W_{dec} maps $V_L \rightarrow \hat{y}$. Modes: **backprop** ($\nabla_W L$ via autograd), **local** ($\Delta W = \eta \langle \delta_0 V_L^T \rangle_B$), **frozen** ($\Delta W = 0$).

Quantity	Symbol	Convention
Voltage	V	Conductance-stage shunting voltage in $[0, 1]$ under nonnegative drive; additive controls may be signed
Conductances	$g^{\text{syn}}, g^{\text{den}}$	Nonnegative via $g = \text{softplus}(\theta)$; local gradients are mapped to raw θ by the transform derivative
Leak conductance	g^{leak}	Set to 1
Input resistance	R^{tot}	≤ 1

Table S7: Units and normalization.

594 Representative Manuscript Architectures

595 Hyperparameters

596 Table S9 collects the core training hyperparameters for each main experiment. Learning rates are
597 applied through parameter groups (top- k , dendritic block-linear, reactivation, decoder) with the values
598 given in the representative config files; where a single LR is reported, every group uses that value.

599 Effective Broadcast Dimensionality in Main Architectures

600 Compute Resources

601 All experiments were run on an institutional GPU cluster. Reproduction runs use one NVIDIA A100
602 or H100-class GPU (40–80GB memory) per seed and do not require distributed training. Table S11
603 gives approximate single-GPU runtime ranges for the reported experiment families; exploratory
604 development runs outside the reported figures used additional cluster time and are not counted in the
605 reproduction estimate.

606 Code, Data, and Asset Availability

607 MNIST, Fashion-MNIST, and CIFAR-10 are public datasets or public benchmark assets cited in the
608 paper; we use them under their standard published access conditions and cite their original sources
609 in the bibliography. To preserve double-blind review, the submission provides code through an
610 anonymized supplementary ZIP rather than a public repository URL. The ZIP contains the model,
611 training, diagnostic, and figure-generation code together with representative configuration files and
612 reproduction instructions for the main experimental families; code and configurations will be de-
613 anonymized after review. The paper does not release a new dataset or a stand-alone pretrained model
614 asset. Dataset access pages are the original MNIST site / UCI entry (<https://archive.ics.uci.edu/dataset/683/mnist+database+of+handwritten+digits>), the Fashion-MNIST reposi-
615 tory (<https://github.com/zalandoresearch/fashion-mnist>), and the CIFAR page (<https://www.cs.toronto.edu/~kriz/cifar.html>).
616
617

618 Algorithm

Algorithm 1 Main 5F Rank-1 LocalCA Update

- 1: **Input:** Dendritic model, batch (x, y) , learning rate η
- 2: Forward pass; loss L , output error δ^y
- 3: Somatic/core error $\delta_0 = \text{decoder-input Jacobian product } J_{\text{dec}}(h_{\text{core}})^\top \delta^y$ (or $W_{\text{dec}}^\top \delta^y$ for a linear decoder)
- 4: **for** each layer n (reverse) **do**
- 5: Rank-1 broadcast $e_n = \text{mean}(\delta_0) \mathbf{1}_n$
- 6: Update EMA estimates for r_n^{4F} and ϕ_n from batch voltages
- 7: $\Delta g_j^{\text{syn}} \leftarrow \eta r_n^{4F} \phi_n \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n \rangle_B$
- 8: $\Delta g_j^{\text{den}} \leftarrow \eta r_n^{4F} \phi_n \langle R_n^{\text{tot}} (V_j - V_n) e_n \rangle_B$
- 9: **end for**
- 10: Optional controls replace line 5 with neuron-wise, rank- K , path-structured, or transported-oracle broadcasts.
- 11: Clip gradients; optimizer step

C Theoretical Details

Random-Broadcast Alignment Intuition

Feedback-alignment arguments suggest that random or low-rank feedback can be useful when the induced local update remains positively correlated with the exact gradient [13]. In this model, that correlation depends on how local eligibility factors co-vary with the conductance-stage path gain in Eq. (4). We treat this as intuition only. The central theoretical object in the paper is still the exact pre-reactivation compartment error $\partial L / \partial V_n$; under identity upward transfer this reduces to $\alpha_n^{\text{cond}} \delta_0$, while post-voltage activation uses $\tilde{\alpha}_n \delta_0$. The empirical question is how faithfully different broadcast fields approximate that quantity.

Morphology-Aware Extensions

Path-integrated propagation. Modulate broadcast error by $\pi_n = \pi_{n-1} \cdot R_{n-1}^{\text{tot}} \cdot \bar{g}_{n-1}^{\text{den}}$, approximating depth attenuation from Eq. (4).

Depth modulation. Per-branch scaling $\kappa_j = \kappa_{\text{base}} / (d_j + d_0)$, mirroring cable attenuation.

Dendritic normalization. $\Delta g_j^{\text{den}} \leftarrow \Delta g_j^{\text{den}} / (\sum_k g_k^{\text{den}} + \varepsilon)$, analogous to homeostatic scaling [25].

Pathway-vector feedback. For tasks with latent pathway structure, the broadcast becomes a role vector inferred from the router. Local pathway activity gates this vector. Upward block transport then passes it to earlier branches, aligning their feedback with downstream descendants.

Router-derived branch roles. Each branch receives a role profile inferred from router assignments and incoming block weights rather than from a hand-coded apical/basal label. Plasticity is then modulated by branch selectivity and a synapse-specific role-alignment factor, emphasizing pathway-consistent updates without imposing a heuristic branch taxonomy.

HSIC Auxiliary Objectives

Following [23], we use kernel-based HSIC objectives:

$$\mathcal{L}^{\text{self}} = B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Z \mathbf{H}), \quad \mathcal{L}^{\text{target}} = -B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Y \mathbf{H}).$$

Moderate weights (0.01–0.1) help on context gating; negligible on MNIST. Online statistics (r_n^{4F} , ϕ_n) use Welford’s algorithm [24].

D Task and Robustness Details

Each synthetic task is generated procedurally from a fixed random seed and uses fixed train/val/test splits for reproducibility. The MNIST-derived synthetic tasks use flattened 28×28 inputs; cue integration instead uses a small vector input with two output classes. Output dimensionality therefore varies by task and is stated in the task description when it differs from 10-way digit classification.

Context gating. Inputs are contextual figure-ground MNIST examples. The right half of the image carries the original MNIST digit, while the left half is replaced by low-amplitude uniform noise; all pixels therefore remain nonnegative. This task tests whether credit assignment can exploit a simple context-dependent spatial structure without leaving the nonnegative-drive regime of the conductance model. The HSIC auxiliary objective (weight 0.01) is used to decorrelate layer representations, adding roughly 3 pp under local learning.

Noise resilience. Inputs are flattened MNIST images corrupted by structured channel noise: a fixed 50-dimensional latent Gaussian perturbation is projected into input space and added to each example, with $\sigma_{\text{task}} = 1.5$, then clipped to $[0, 1]$. The projection is fixed across training, so the network must filter correlated interference to recover the underlying class signal. Credit assignment must remain coherent under the corrupted signal, making this task particularly sensitive to path-gain dispersion.

661 **Cue integration.** Two noisy cue streams (A and B) are presented simultaneously, each carrying
662 partial information about one of 2 classes. A binary one-hot context signal indicates which cue is
663 reliable on each trial (cue A or cue B), and each cue vector is clipped to $[0, 1]$ after noise is added.
664 The fixed-pathway variant duplicates the context into both cue branches; the learned-routing variant
665 requires the router to discover the cue separation from data. This task is used in Sec. A to test
666 structured pathway-vector broadcast while staying inside the nonnegative-input regime.

667 **Depth Scaling and Noise Robustness**

668 **Depth scaling.** Varying dendritic depth from 1–4 layers (branch factors $[9]$ to $[3, 3, 3, 3]$): shunting
669 local degrades from 63.5% to 57.4%; additive local degrades more steeply from 54.9% to 29.7%.
670 The shunting advantage grows with depth ($+8.5 \rightarrow +27.7$ pp). Backprop ceilings remain stable at
671 about 90–92%. See Fig. S5A.

672 **Noise robustness.** Gaussian noise $\mathcal{N}(0, \sigma^2)$ on broadcast error: shunting is robust to $\sigma \leq 0.1$ (about
673 62%) while additive drops from 46.5% to chance at $\sigma=1.0$, confirming that shunting credit signals
674 carry useful learning information beyond the broadcast magnitude. See Fig. S5B.

Setting	Encoder	E layers	Tree	Synapses / branch	Seeds	Notes
Gradient fidelity / path-gain / noise-resilience mechanism	id.+ReLU	[128, 128]	[3, 3]	$N_E=40, N_I \in \{0, 5, 10, 20, 40\}$	5 fidelity, 5 oracle	Main mechanism sweeps in Figs. 2 and 3; reactivation <code>param_tanh</code> .
MNIST / context-gating local competence	id.+ReLU	[128]	[3, 3]	$N_E=40, N_I=20$	5	Best local 5F per-soma runs in Table S2; reactivation <code>param_tanh</code> .
Fashion-MNIST competence	id.+ReLU	[128]	[3, 3]	$N_E=40, N_I=20$	5	Dedicated five-seed standard-vs-local competence sweep; reactivation <code>param_tanh</code> .
Cue-routing PV-LocalCA	router	[64]	[2]	$N_E=10, N_I=4$	5	Learned router with two pathway groups and rank-2 structured feedback; reactivation <code>param_tanh</code> .
CIFAR-10 compact direct-I-stream control ladder	id.+ReLU	[20] E, no I cells	[3, 3, 3, 3]	$N_E=25, N_I=25$ direct I-to-E, with matched no-I controls	5 per condition	Harder-data extension with input-driven inhibitory conductance, decoder [32, 16], decoder-aware soma mapping, and additive fairness controls.
Morphology $\times$ inhibition appendix map	id.+ReLU	[128, 128]	[2, 2], [3, 3], [4, 4], [2, 2, 2], [3, 3, 3]	$N_E=40, N_I \in \{0, 5, 10, 20, 40\}$	3	Supportive 3-seed sweep; reactivation <code>param_tanh</code> .

Table S8: Representative architectures used in the manuscript. All dendritic rows use explicit branch-level excitatory and inhibitory synapse banks without direct somatic synapses, nonnegative conductances, and a nonnegative first-layer drive enforced by an explicit ReLU transfer stage in the main runs. The tree column gives dendritic morphology per excitatory neuron, and the synapse counts report excitatory and inhibitory synapses per branch.

Setting	Opt.	LR	Batch	Epochs	W-dec.	Notes
MNIST / Fashion-MNIST / context gating (5F per-soma LocalCA)	Adam	$10^{-3} / 5 \cdot 10^{-4}$ (block / react.)	256	100	0	fixed-epoch, no early stopping; <code>param_tanh</code> reactivation
MNIST / Fashion-MNIST / context gating (BP ceiling)	Adam	$10^{-3} / 5 \cdot 10^{-4}$	256	100	0	matched to LocalCA setup
Gradient fidelity / path-gain / noise resilience	Adam	10^{-3}	256	50	0	5 seeds, hooks enabled for diagnostic capture
Morphology $\times$ inhibition regime map (supportive)	Adam	10^{-3}	256	50	0	3 seeds
Cue routing	Adam	10^{-3}	256	90	0	early stopping with patience 30
CIFAR-10 compact direct-I-stream depth-4 LocalCA	Adam	$10^{-3} / 10^{-4}$ (block / react.)	256	400	0 / 0.01	grad-clip 5.0, early stop patience 50, nonlinear decoder
CIFAR-10 compact direct-I-stream depth-4 BP	Adam	10^{-3}	256	200	0.01	early stop patience 40, matched ceiling

Table S9: Training hyperparameters by experiment. All runs use Adam with default $\beta_1=0.9$, $\beta_2=0.999$, $\varepsilon=10^{-8}$. Reactivation and block-linear parameter groups use a slower LR to keep conductance-level dynamics stable. Representative configs are provided in the anonymized supplementary ZIP.

Setting	Hidden width(s)	Output dim	Effective default per-soma field
Gradient fidelity / path-gain / noise resilience	128, 128	10	Scalar / rank-1 at both dendritic layers
MNIST / Fashion-MNIST competence	128	10	Scalar / rank-1 at the dendritic hidden layer
Context gating competence	128	2	Scalar / rank-1 at the dendritic hidden layer
Cue routing	64	2	Scalar / rank-1 by default; higher-rank tests use explicit low-rank or pathway-vector feedback
Compact CIFAR-10 harder-data extension	20	decoder-input dim 20	Vector-valued per-soma after decoder-aware mapping; higher-rank tests use explicit low-rank or path transport

Table S10: **Effective dimensionality of the default neuron-wise broadcast in the main architectures.** The implementation uses the output error vector only when layer width matches decoder output dimension; otherwise it falls back to a shared scalar. The main feedforward settings in this paper therefore test a genuinely rank-1 broadcast regime by default, which is why the higher-rank low-rank, path-transport, and pathway-vector controls are important.

Experiment family	Runs	Time / run	Approx. GPU-h
Competence and verification	~25	5–20 min	4–9
Gradient, path-gain, inhibition, and oracle diagnostics	~80	10–30 min	15–40
Morphology, stress, rule, and feedback controls	~120	5–35 min	20–70
CIFAR-10 control ladder	70	0.8–2.5 h	60–175
Figure generation and CPU-side summaries	–	< 5 h total	< 5

Table S11: **Approximate compute for reported results.** Ranges reflect single-seed wall-clock variation across A100/H100-class GPUs, data-loading overhead, and early stopping.

Asset	Use in this paper	Access / license or terms note
MNIST	Digit classification and derived nonnegative synthetic tasks	Public benchmark from the original MNIST site / UCI entry; UCI lists DOI 10.24432/C53K8Q and asks users to follow the original acknowledgement policy; no raw-data redistribution.
Fashion-MNIST	Apparel classification stress test	Public Zalando Research benchmark; repository is MIT licensed; no raw-data redistribution beyond standard dataset loaders.
CIFAR-10	Flattened harder-data diagnostic	Public CIFAR dataset site asks users to cite Krizhevsky’s technical report; no raw-data redistribution.
Synthetic tasks	Context gating, noise resilience, cue integration	Generated procedurally from public benchmarks or random seeds described in Appendix D; no new third-party asset is introduced.

Table S12: **Existing assets used in the manuscript.** We credit the original sources in the bibliography, use public benchmark assets under their published access conditions, and do not redistribute third-party raw data in the anonymous submission.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction state the paper’s main mechanistic claim, the scope of the conductance-based setting, and the regime-dependent nature of the empirical results; see Sec. 1 and Sec. 5.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: The discussion explicitly narrows the scope of the claims and discusses limitations involving benchmark strength, broadcast assumptions, depth scaling, morphology, activation shape, and feedback rank; see Sec. 5.

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- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
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